UE23CS320A - Project Phase 2 Review

A Deep Learning and Generative AI-Driven Diet Recommendation System for Personalized Chronic Disease Management

Updated review deck using the same ISA-1 headings with refreshed dataset counts, source sites, benchmark comparison, full-train model status, and terminal demo steps.

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Prepared from the repository state on April 19, 2026.

Problem Statement

The project still targets safe, adaptive nutrition support, but Phase 2 focuses on making the disease-risk layer reliable first.

Clinical problem

- Generic diet tools ignore disease-specific risk and patient state.
- Unsafe recommendations can happen if prediction and nutrition layers drift apart.
- The system must explain risk before it can justify meal correction.

Implementation problem

- Merged survey datasets can look large but still lose critical lab and blood-pressure columns.
- Terminal and backend demos break if model artifacts are too heavy or feature schemas are inconsistent.
- The review deck therefore needs current numbers, not earlier placeholders.

Phase 2 answer

- Retrain project-final disease models on the large prepared splits.
- Keep benchmark comparison visible against baseline models.
- Make terminal prediction reproducible from the actual deployed artifacts.

Current implementation priority: stable disease prediction first, diet calibration next.

Abstract and Scope

The scope is now a working disease-risk backbone that cleanly feeds the next meal recommendation step.

In scope now

- CKD, hypertension, and diabetes risk modeling.
- Preprocessing, training, benchmarking, threshold building, and terminal/API inference.
- Dataset audit notes, source tracking, and screenshot-ready demo commands.

Next scope after this review

- Raw patient-friendly input mapping into each disease feature space.
- Direct coupling of risk levels into the diet recommendation engine.
- End-to-end meal recalibration once the raw-input layer is finalized.

This keeps the original project vision but narrows the live demo to what is fully implemented now.

Suggestions from Phase 1

The Phase 1 review directly influenced the current implementation decisions.

Panel suggestions

- Do not depend only on synthetic or weak placeholder data.
- Avoid spreading effort across too many disconnected ideas.
- Prioritize the most practical models and flows for the review demo.

What changed

- The project retained large amalgamated disease tables but audited them instead of trusting them blindly.
- The final artifacts were retrained with lighter, deployable models.
- The terminal demo inputs were updated to match the active feature schemas.

Result

- Project-final models now load and predict in terminal.
- Training uses full 400k+ prepared splits per disease.
- The review deck can now cite exact row counts, columns, and source sites.

The present repo state reflects those review-driven corrections.

Requirement Specification

The current milestone satisfies the core risk-layer requirements needed for the next implementation step.

Functional requirements covered

- Disease-wise preprocessing and train/test generation.
- Model artifact creation for CKD, hypertension, and diabetes.
- Terminal prediction from JSON input and template generation.
- Baseline comparison and threshold/factor reports.

Non-functional requirements covered

- Repeatable commands for review screenshots.
- Smaller artifacts that can actually load for demo use.
- Clear source-site traceability for the amalgamated data.
- Explicit separation between current backend work and next-step diet calibration.

Current review assets now live under outputs/isa-review2 and reports/current.

Confirmed Datasets

The project is an amalgamation of disease, public-health, and nutrition sources.

Disease and public-health sources

- UCI Machine Learning Repository
- Framingham Heart Study
- Pima Indians Diabetes Dataset
- CDC BRFSS Annual Data
- CDC NHANES 2017-2018

Nutrition sources

- ICMR-NIN IFCT 2017
- Mendeley Data
- Open Food Facts India
- Kaggle support files where relevant

What is confirmed here

- Site names for the full data amalgamation are explicitly listed.
- Dataset row counts and column names are now documented per active serving dataset.
- Large train splits exceed 400k rows for every disease model.

Source sites used across the project: UCI Machine Learning Repository | Framingham Heart Study | Pima Indians Diabetes Dataset | CDC BRFSS Annual Data | CDC NHANES 2017-2018 | ICMR-NIN IFCT 2017 | Mendeley Data | Open Food Facts India | Kaggle.

Temporary Datasets

Dataset detail for CKD and Hypertension.

CKD large serving dataset

File: medical datasets/large/ckd_large.csv
Rows: 455,590 | Columns: 10
Training rows used for final model: 690,956
Target: ckd_label
Final model: DecisionTreeClassifier

Columns:
age, male, BMI, sysBP, diaBP, glucose, diabetes, prevalentHyp, source, ckd_label

Sites used in amalgamation:
CDC BRFSS Annual Data
CDC NHANES 2017-2018
UCI Machine Learning Repository

Hypertension large serving dataset

File: medical datasets/large/hypertension_large_500k.csv
Rows: 500,000 | Columns: 13
Training rows used for final model: 785,242
Target: prevalentHyp
Final model: LogisticRegression

Columns:
male, age, education, currentSmoker, cigsPerDay, prevalentStroke, BMI, sysBP, diaBP, glucose, diabetes, source, prevalentHyp

Sites used in amalgamation:
CDC BRFSS Annual Data
Framingham Heart Study

These cards include rows, columns, training rows used, model used, column names, and source-site names.

Temporary Datasets

Dataset detail for Diabetes plus the current 400k+ training proof.

Diabetes large serving dataset

File: medical datasets/large/diabetes_large_500k.csv
Rows: 500,000 | Columns: 10
Training rows used for final model: 754,970
Target: diabetes
Final model: GaussianNB

Columns:
age, male, BMI, sysBP, diaBP, glucose, prevalentHyp, currentSmoker, source, diabetes

Sites used in amalgamation:
CDC BRFSS Annual Data
Pima Indians Diabetes Dataset
Kaggle

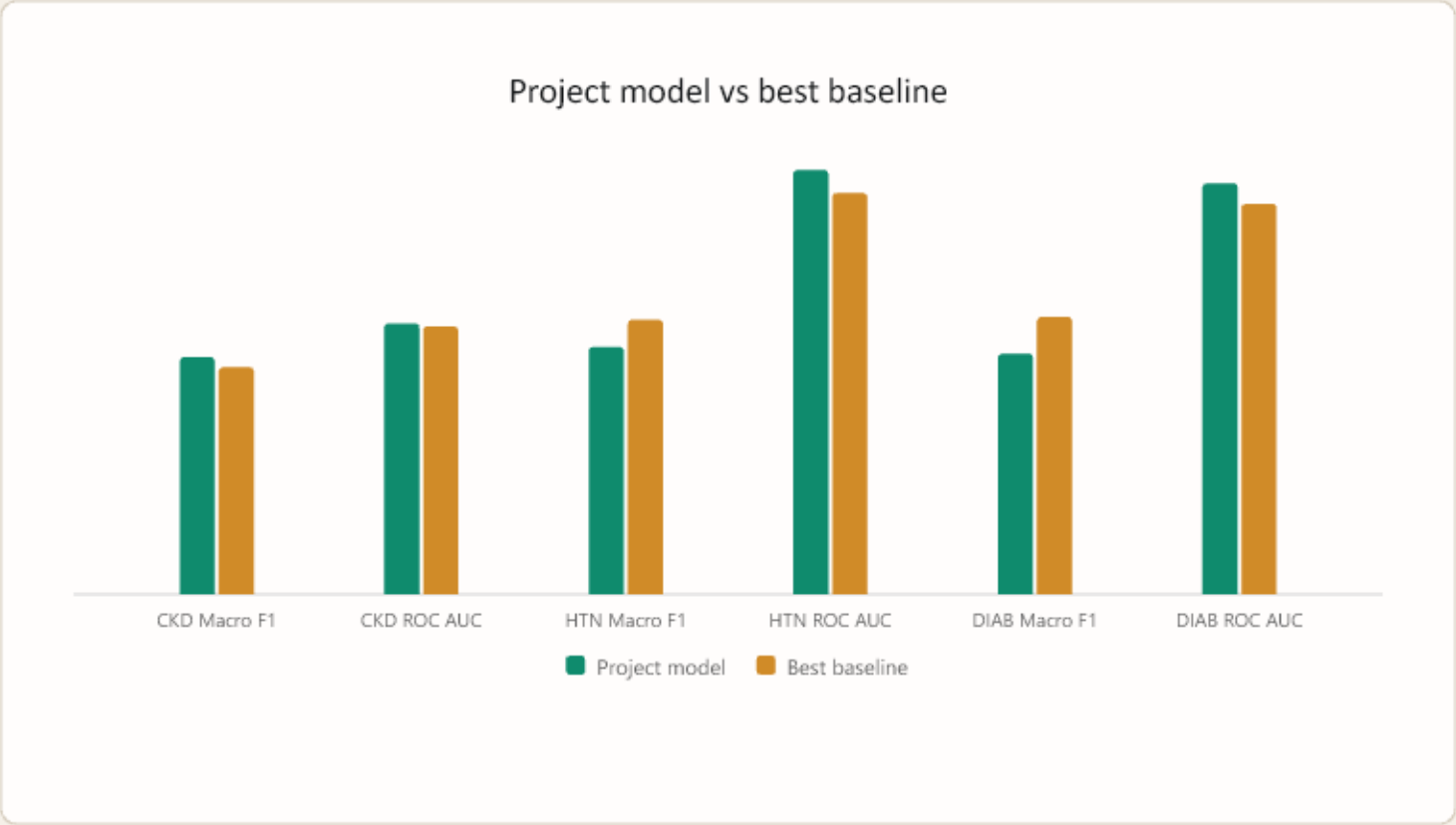
Training proof and caveat

- CKD train split used for the final model: 690,956 rows.
- Hypertension train split used for the final model: 785,242 rows.
- Diabetes train split used for the final model: 754,970 rows.
- Baseline benchmarking remains capped at 120,000 rows per disease for runtime control.
- That cap only applies to the comparison study, not to the saved project-final artifacts.

This slide answers the panel request to show that the deployed models were trained on at least 400k values per disease.

Temporary Datasets

Model comparison graph and the exact terminal commands needed for screenshots.



Commands for training + comparison screenshot

```
Set-Location "D:\sem 6\capstone\Sem_6_Capstone"
& ".\.venv\Scripts\python.exe" scripts\train_models.py --data-source
preprocessed
& ".\.venv\Scripts\python.exe" scripts\build_risk_profiles.py
& ".\.venv\Scripts\python.exe" scripts\benchmark_baseline_models.py
```

Commands for terminal prediction screenshot

```
& ".\.venv\Scripts\python.exe" scripts\predict_from_terminal.py --disease
ckd --show-template
& ".\.venv\Scripts\python.exe" scripts\predict_from_terminal.py --input
demo_inputs\terminal_samples\ckd_sample_1.json
& ".\.venv\Scripts\python.exe" scripts\predict_from_terminal.py --input
demo_inputs\terminal_samples\hypertension_sample_1.json
& ".\.venv\Scripts\python.exe" scripts\predict_from_terminal.py --input
demo_inputs\terminal_samples\diabetes_sample_1.json
```

Chart note: project metrics come from full-train artifacts; best baseline metrics come from the capped benchmark report.

Design Approach

The implementation keeps the original layered design but tightens the modeling core.

Approach followed

- Use prepared disease-specific large tables for training.
- Store final artifacts separately from baseline comparison outputs.
- Keep the meal recommendation layer downstream of verified risk inference.

Why this approach

- The review demo needs a working prediction path before a richer meal-calibration path.
- A lighter model family makes deployment and terminal inference feasible.
- The same headings can be preserved while the content becomes implementation-driven.

Current outcome

- CKD final model: DecisionTreeClassifier
- Hypertension final model: LogisticRegression
- Diabetes final model: GaussianNB

The model family is intentionally lighter than the earlier Random Forest artifacts to keep inference practical.

Design Constraints, Assumptions & Dependencies

Current constraints are mostly about data realism, deployability, and review-time clarity.

Constraints

- The deck must reflect the exact repo outputs as of April 19, 2026.
- Terminal samples must match the deployed feature schema.
- Large-data training is required, but baseline comparison must still finish in reasonable time.

Assumptions

- The prepared large splits remain the active training source for this review snapshot.
- Baseline comparisons are acceptable on a capped training subset as long as that is stated clearly.
- The next coding step will continue from the saved artifacts already produced here.

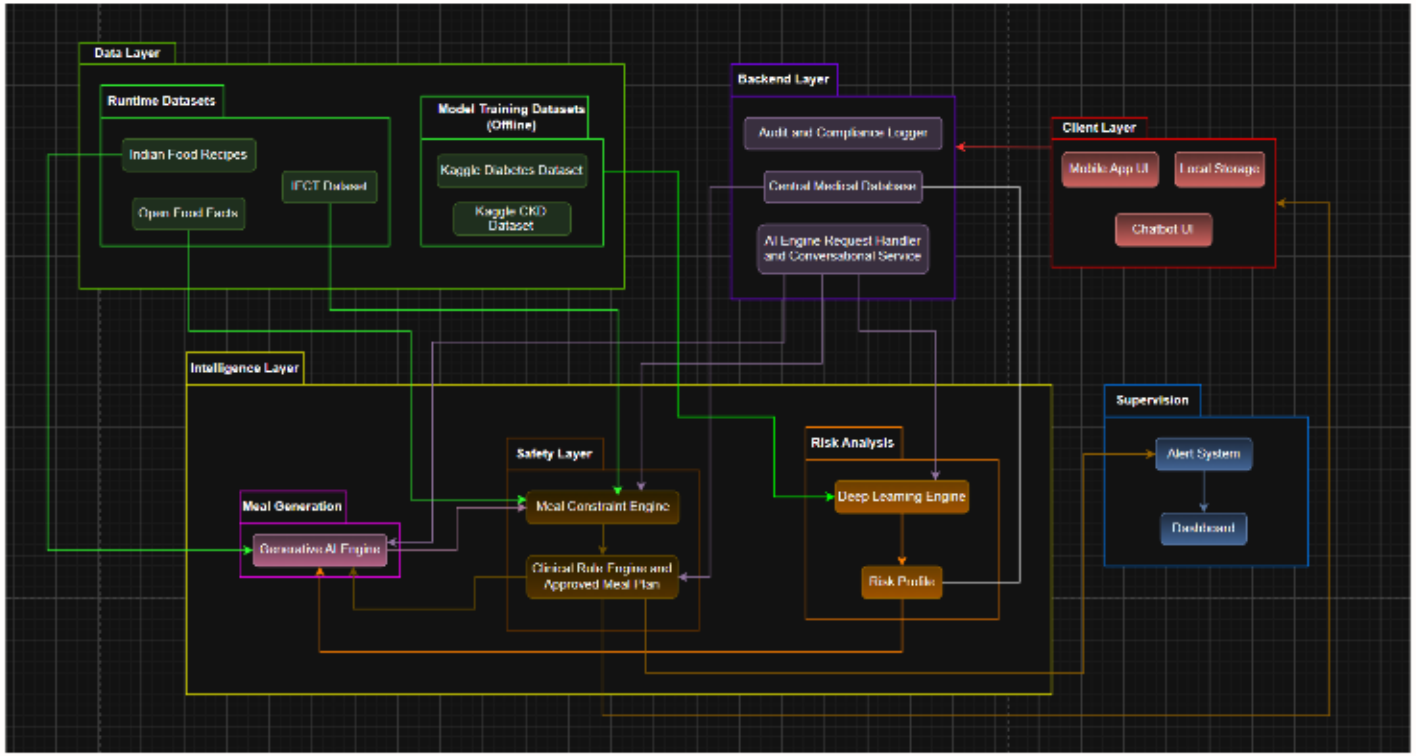
Dependencies

- Python + scikit-learn + imbalanced-learn
- Saved artifacts under artifacts/
- Benchmark and threshold reports under reports/current/
- Legacy diagram images for design-alignment slides

This makes the implementation state transparent without changing the original review structure.

Architecture Diagram

The architecture slide is preserved so the panel can compare design intent with the now-working model/report path.



Implemented now: preprocessing, final-model training, threshold report generation, terminal inference, and benchmark reporting.

Master Class Diagram

The original class-level view is retained for design continuity.

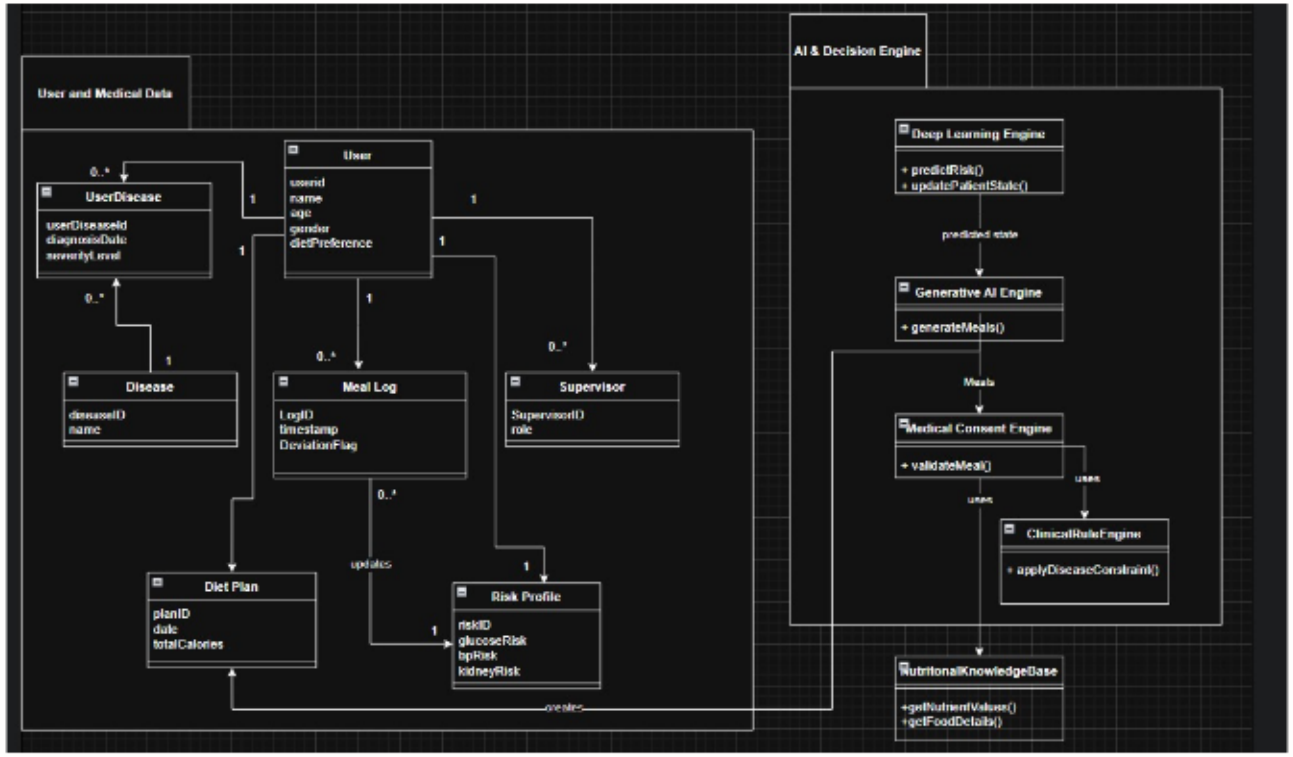


Diagram asset reused from the ISA-1 review material.

ER Diagram

This remains useful for explaining how user, disease, meal, and report data fit together.

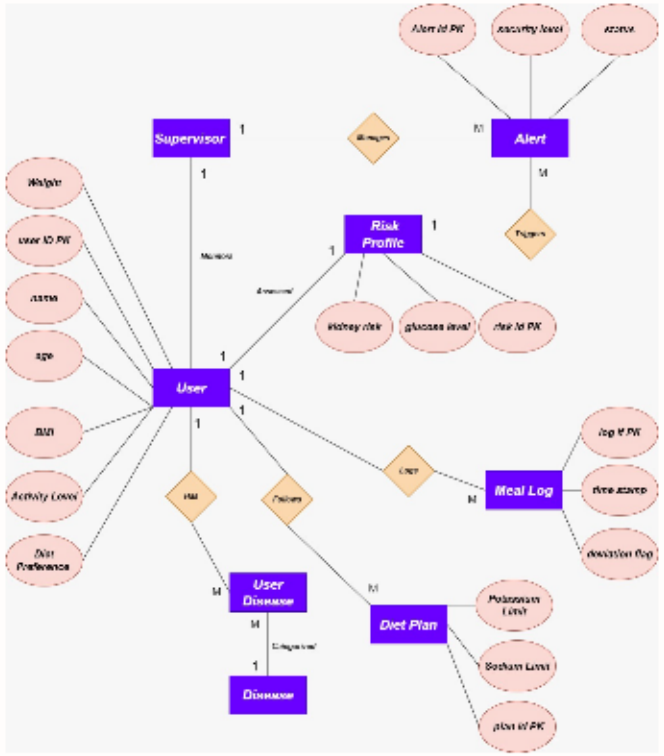


Diagram asset reused from the ISA-1 review material.

Use-Case Diagram

The use-case view still reflects the intended journey once risk prediction connects into meal calibration.

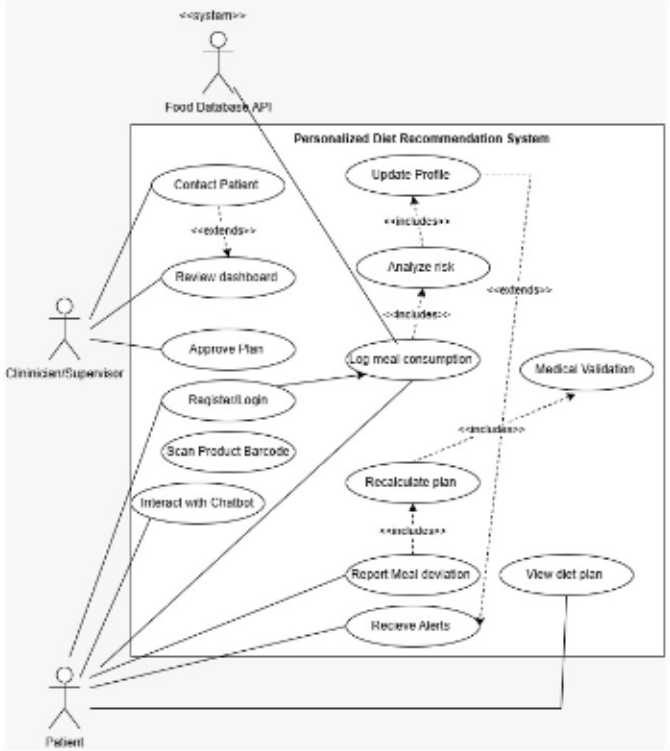
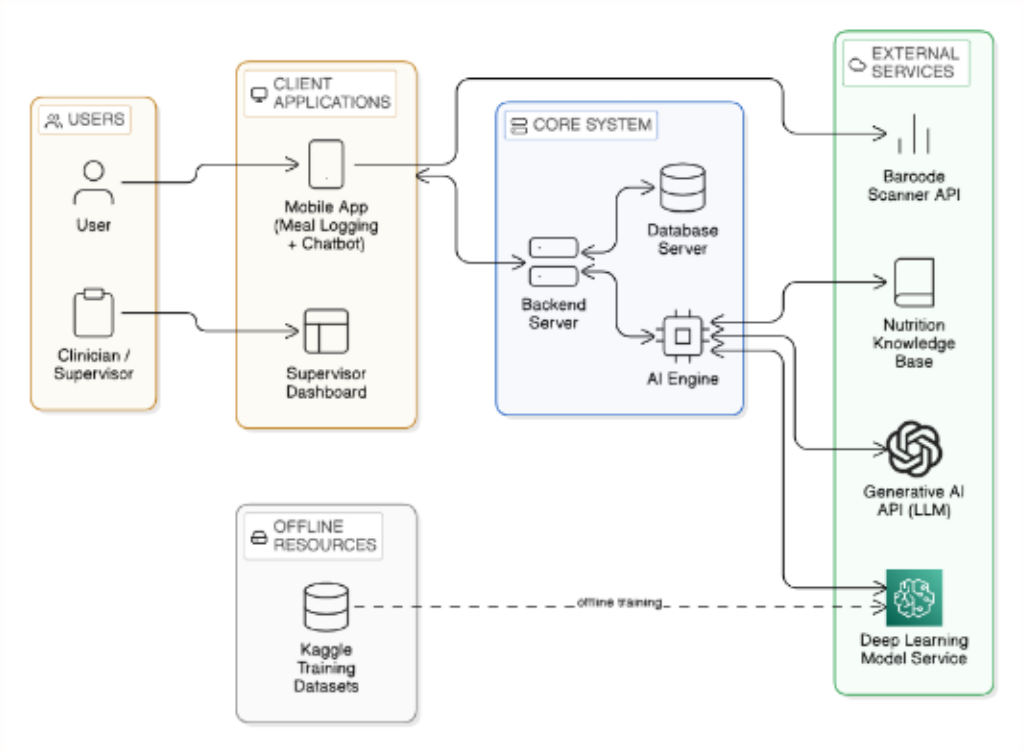


Diagram asset reused from the ISA-1 review material.

External Interface Diagram

For this review, the clearest external interface is the working terminal and backend prediction path.



Terminal prediction now runs using the current lightweight deployed artifacts and schema-correct demo payloads.

Thank You

Current status:
- project-final models retrained
- terminal prediction working
- screenshot commands documented
- comparison graph added
- dataset rows, columns, and source sites documented

Next step: continue the implementation with raw-input mapping and then reconnect risk into diet calibration.

Final models
CKD: DecisionTree
HTN: LogisticRegression
DIAB: GaussianNB

Full-train rows
690,956 | 785,242 | 754,970

Deck follows the exact ISA-1 heading sequence extracted from the legacy review PDF.